

DevOps for Data

Introduction to DevOps, Data Engineering, and
Cloud

Introduction

About me



- EngD 2021-2023 generation
- 10 years+ experience as software engineer
- Work as DevOps engineer at Rabobank

Course Planning

Day One: (DevOps)

Module One: Cloud, Access Management

Module Two: Azure Function, FinOps, Pipeline

Day Two: (Data)

Module Three: Data Factory, Storage

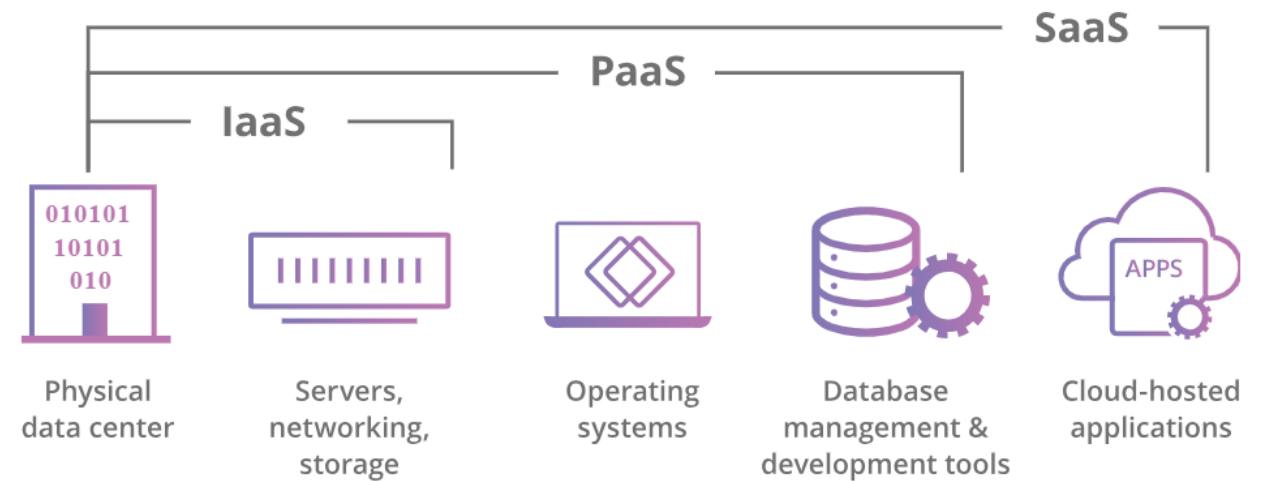
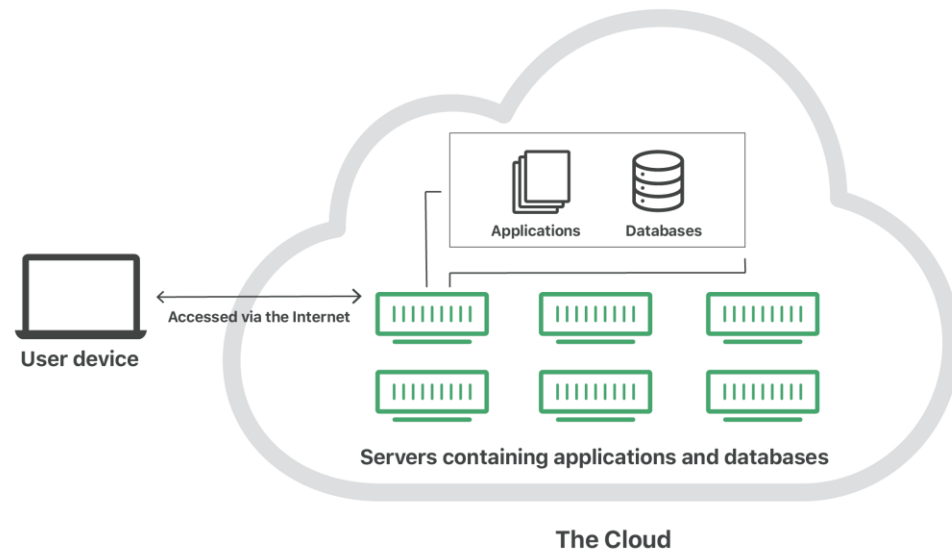
Module Four: Hackathon! (Winners will get an attractive price)

Expectation Management

- Why is this workshop happening?
- We will go through Azure Cloud basics and Data related functionality
- Do things hands on
- Make mistakes and learn from them

Azure Cloud

What is Cloud?



Azure Basics

Azure Account (Entra ID)

The Azure account is a global unique entity that gets you access to Azure services and your Azure subscriptions.

AAD Group

Azure Active Directory which contains some the users.

Subscriptions

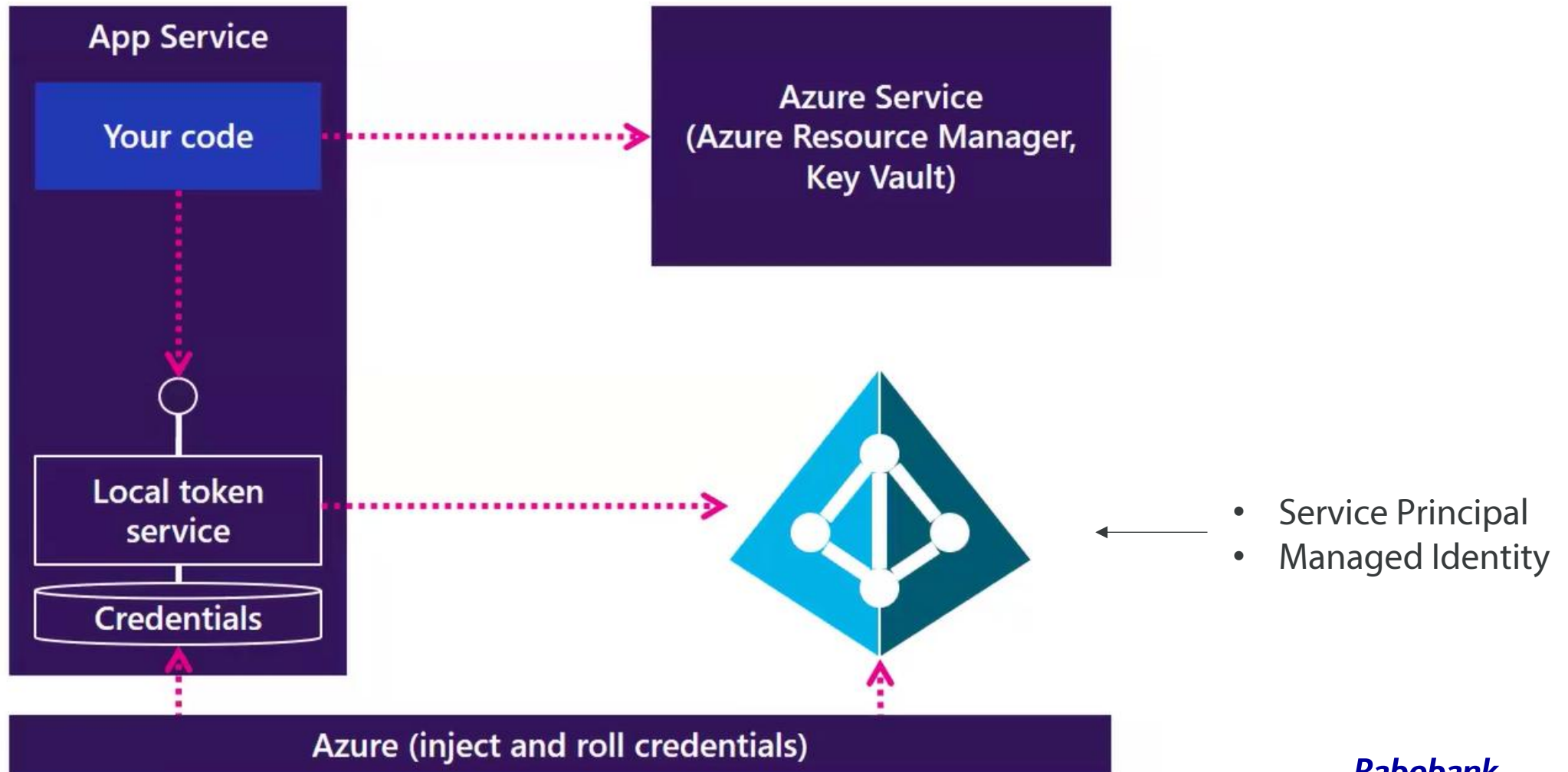
You can create multiple subscriptions in your Azure account to create separation e.g. for billing or management purposes.

Resource Group

Shelves where you keep resources (ie functions, webapps, database).

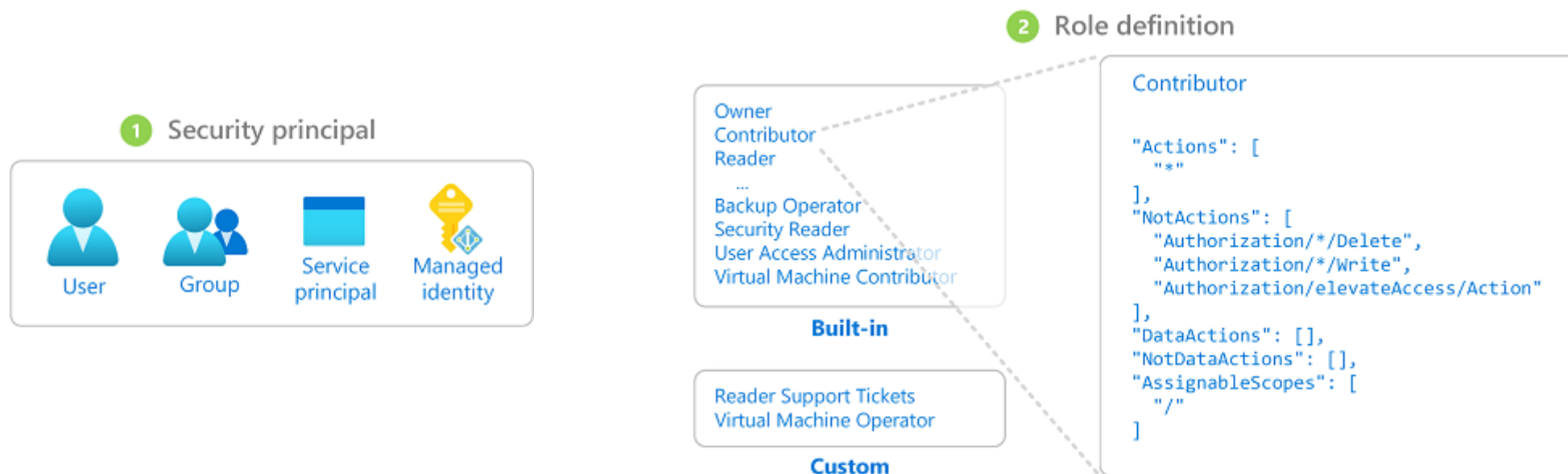
Access Management

Access Control (IAM)



RBAC

Azure role-based access control (Azure RBAC) helps you manage who has access to Azure resources, what they can do with those resources, and what areas they have access to.



Task One: Create Resource Group

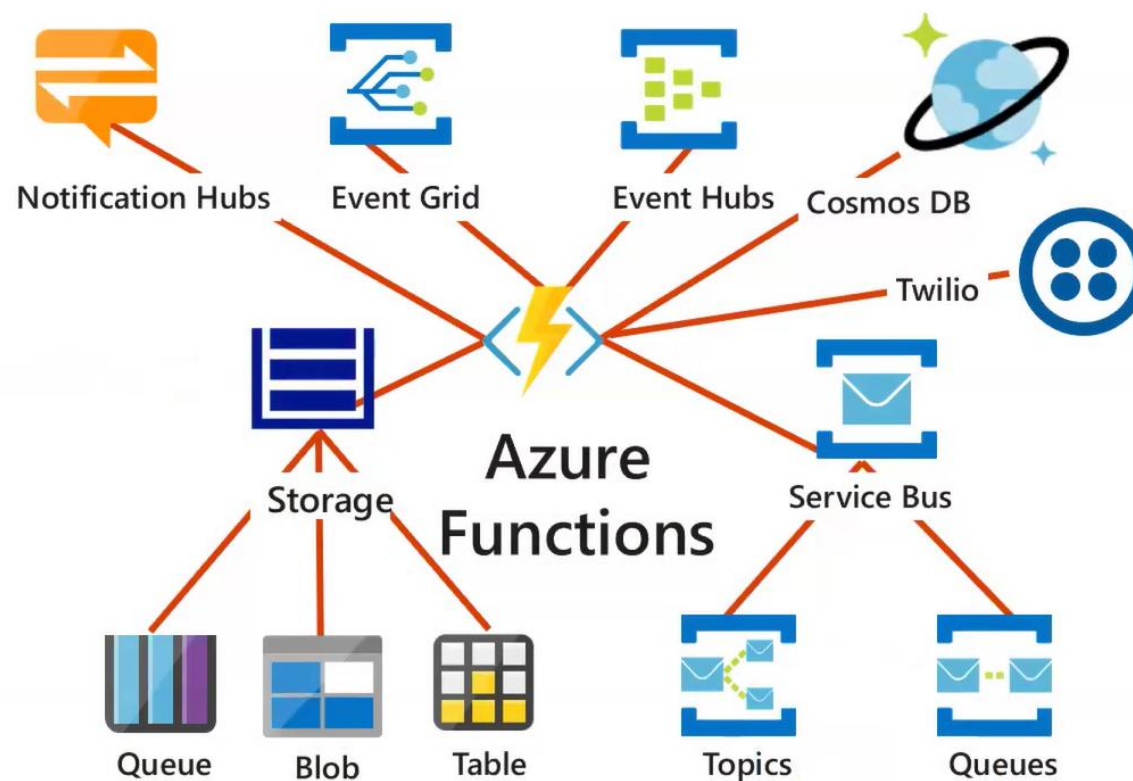
- Go to <https://portal.azure.com/>
- Go to Marketplace > Search Resource Group > Create a Resource Group
- Naming convention: rg-engd-westeu-<team ID> (ie rg-engd-westeu-001)
- One resource group per team

Azure Function

Azure Function

Overview

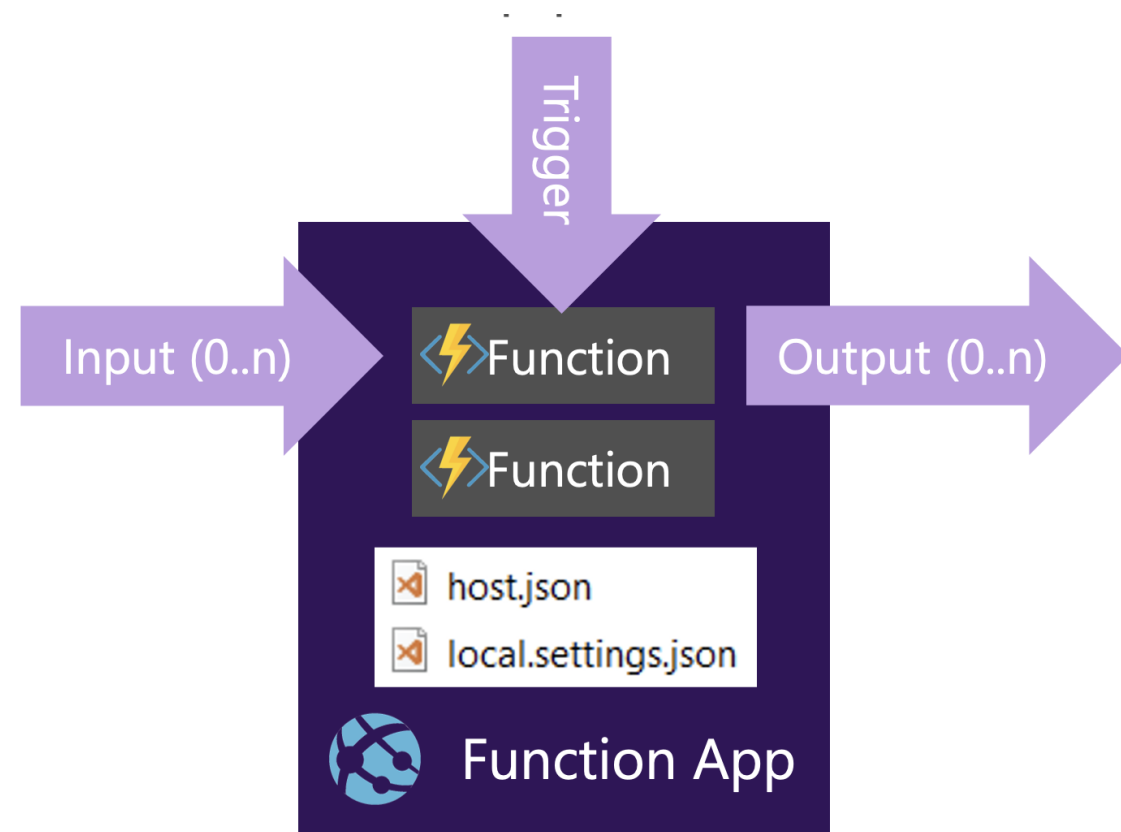
- Consider Functions for tasks like image or order processing, file maintenance, or for any tasks that you want to run on a schedule.
- Azure Functions supports *triggers*, which start execution of your code, and *bindings*, which simplify coding for input and output data.



Triggering Azure Function

Triggers

Service	Trigger description
Blob Storage	Starts a function when a new or updated blob is detected.
Azure Cosmos DB	Start a function when inserts and updates are detected.
Event Grid	Starts a function when an event is received from Event Grid.
HTTP	Starts a function with an HTTP request.
Microsoft Graph Events	Starts a function in response to an incoming webhook from the Microsoft Graph. Each instance of this trigger can react to one Microsoft Graph resource type.
Queue Storage	Starts a function when a new item is received on a queue. The queue message is provided as input to the function.
Service Bus	Starts a function in response to messages from a Service Bus queue.
Timer	Starts a function on a schedule.



How Azure Functions Work

Serverless Function

- Solution for running small pieces of code, or "functions," in the cloud:
 - Write only code that is relevant to business logic
 - Removes the necessity to write "plumbing" code to connect or host application components
- Build on open-source WebJobs code
- Supports a wide variety of programming languages, for instance:



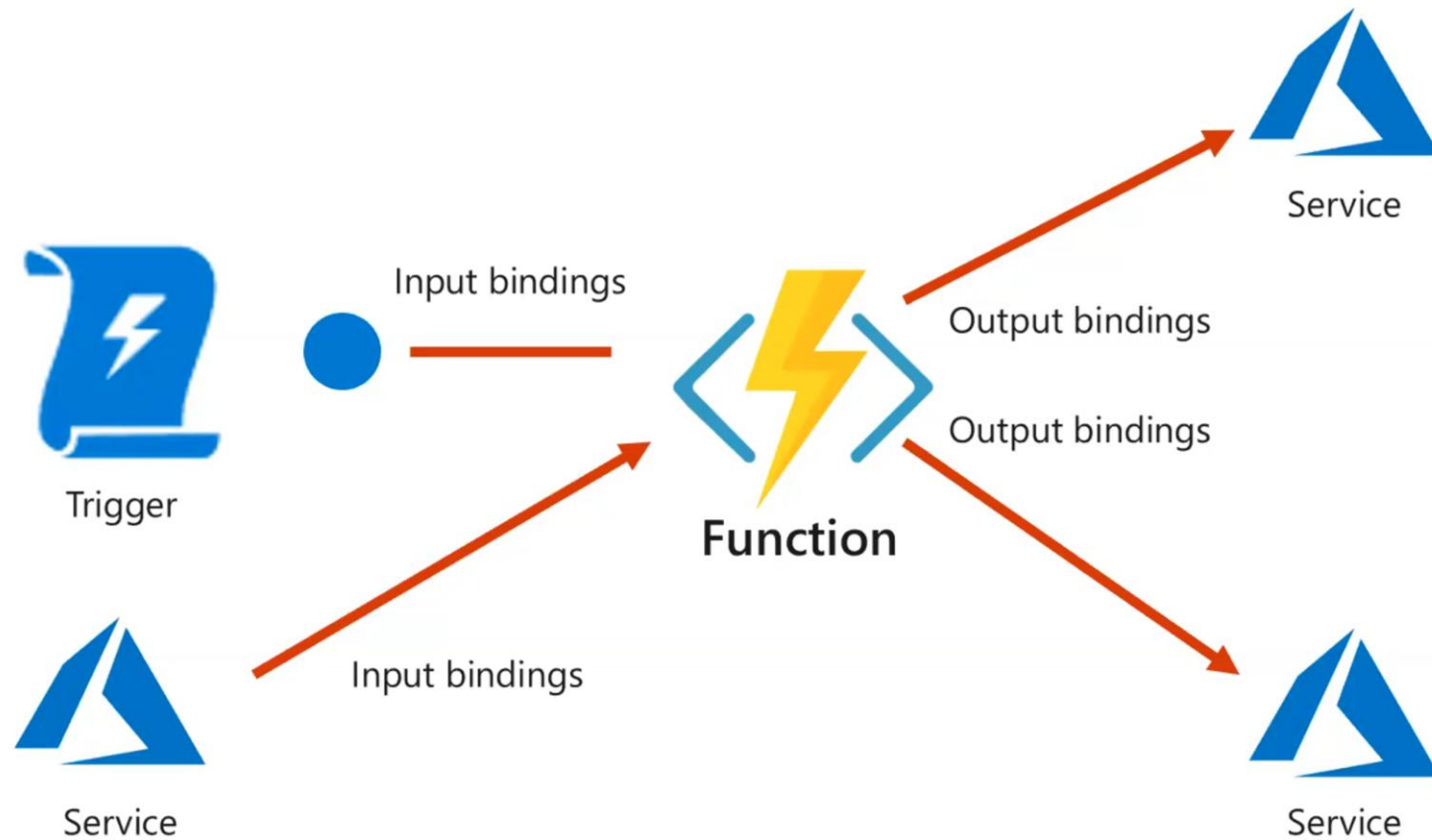
- Even supports scripting languages, such as:



Scaling Azure Function

- The unit of scale for Azure Functions is the function app.
- A *scale controller* monitors the rate of events to determine whether to scale out or scale in.
- Instances may be "scaled in" to zero when no functions are running within a function app.
- The next request has the latency – a *cold start* – of scaling from zero to one

Input and Output Binding



Task Two: Create an Azure Function

- Go to <https://portal.azure.com/>
- Go to Marketplace > Create a Storage Account
- -- Naming convention: **storageengdw**eu<team ID> (ie storageengdw001)
- Go to Marketplace > Search Azure Function> Create an Azure Function
- Naming convention: func-engd-westeu-<team ID> (ie func-engd-westeu-001)
- One function per team
- Create a Http Function inside func-engd-westeu-<team ID>
- Test it
- (Bonus) Create Storage Triggered function

Pipeline

Samples

ETL (Python) Pipeline

```
import json

def read_data(file_path):
    with open(file_path, 'r') as f:
        return f.read()

def write_data(file_path, data):
    with open(file_path, 'w') as f:
        f.write(data)

def apply_transformation(data, transformation_type):
    if transformation_type == 'uppercase':
        return data.upper()
    # Add more transformations as needed
    return data

def main():
    # Load configuration
    with open('config.json', 'r') as f:
        config = json.load(f)

    source_file = config['source_file']
    destination_file = config['destination_file']
    transformation = config['transformation']

    # Read data from source
    data = read_data(source_file)

    # Apply transformation
    transformed_data = apply_transformation(data, transformation)

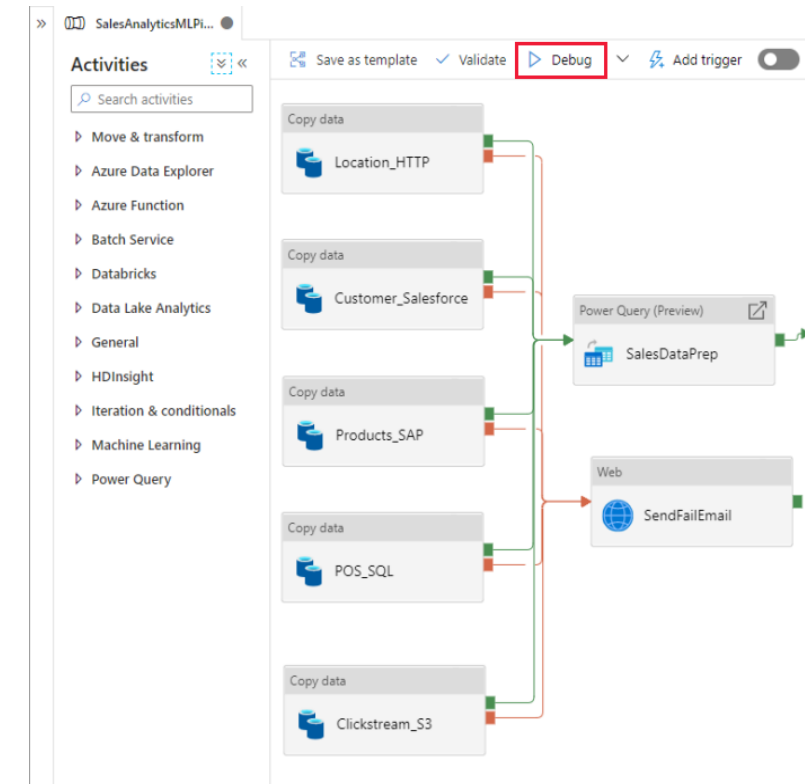
    # Write transformed data to destination
    write_data(destination_file, transformed_data)

if __name__ == "__main__":
    main()
```

GitHub deployment Pipeline

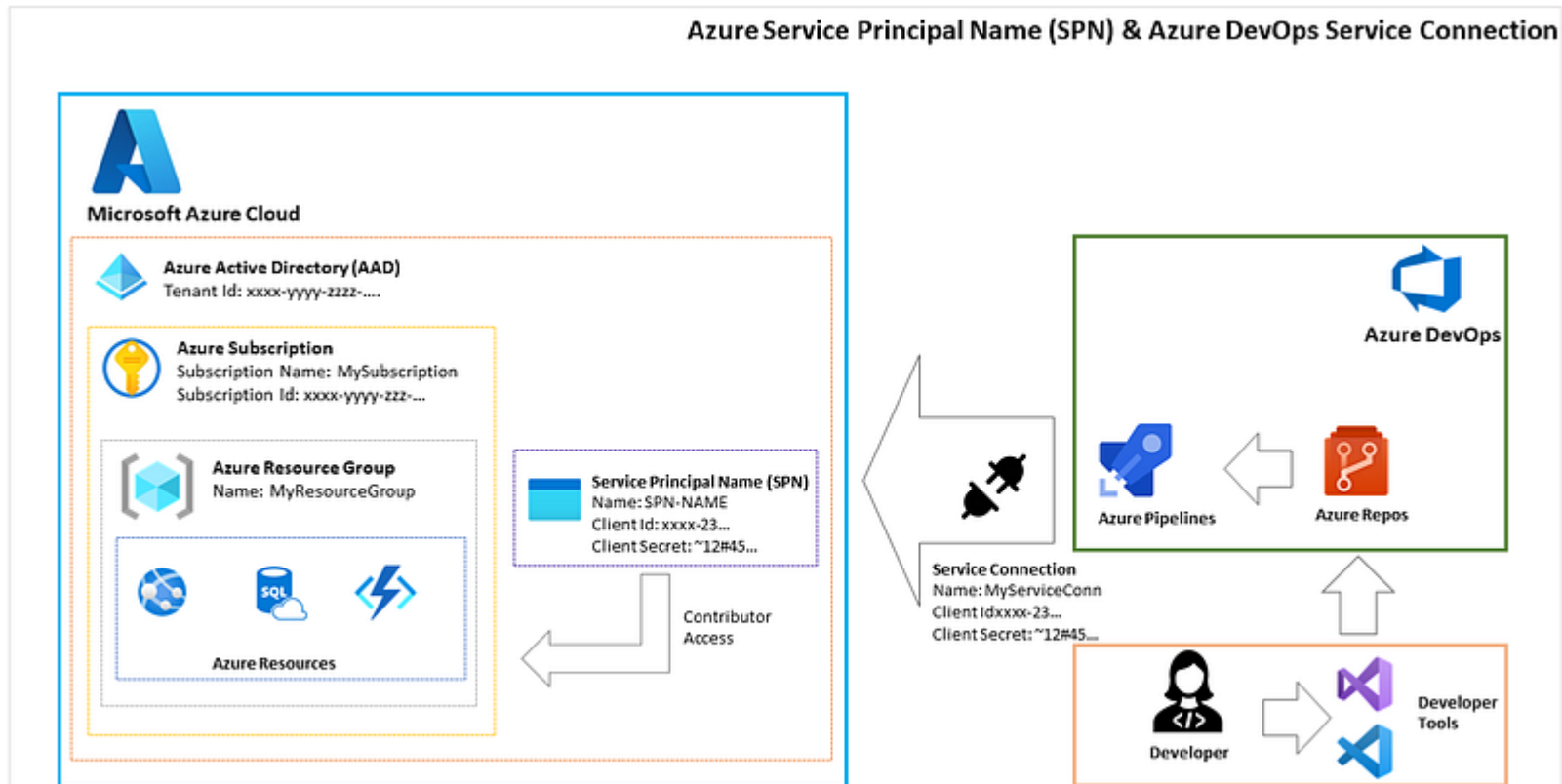
```
1 resources:
2   pipelines:
3     - pipeline: smoke_test
4       source: resi-iot-function-app-Test -Vp
5       trigger:
6         branches:
7           include:
8             - 'azure-pipelines-test-automation'
9
10
11 pool:
12   vmImage: windows-latest
13
14 #Installing the Python
15
16 steps:
17   Settings
18   - task: UsePythonVersion@0
19     inputs:
20       versionSpec: '3.6'
```

Azure Data Factory Pipeline



Infra Pipeline Usage

- In Organizations, Pipelines are used for everything.
- Use pipelines to create resources/infrastructure.
- Service Principal Name (SPN) is used to deploy resources in Azure



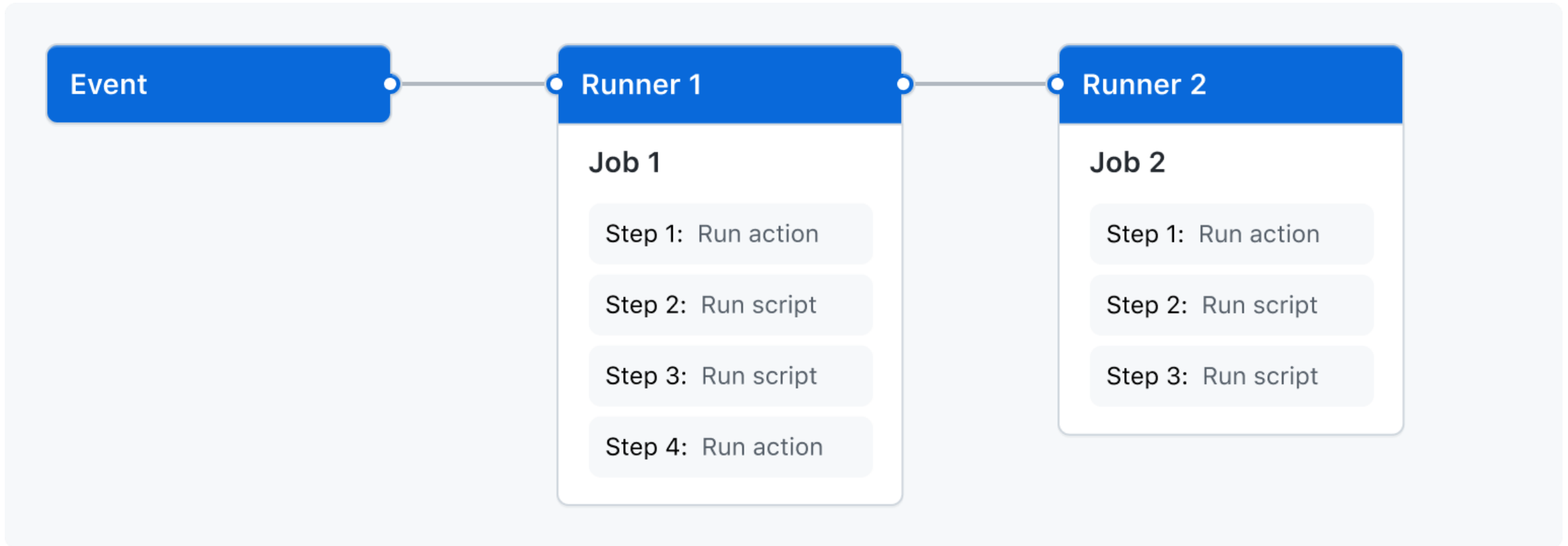
CI/CD Pipeline

- CI/CD Pipeline is used to deploy applications in Infrastructure.
- CI/CD triggers can be:
 - Git events (Git Push, PR merge)
- Best practices include:
 - Unit Testing
 - Code Coverage
 - Security Checks (CodeQL)
 - Code Quality Checks (SonarQube)
 - Packaging (create Artifacts)
 - Storing Packages (Push to ACR/Nexus)
 - Deployment Slots

GitHub Actions

CI/CD Pipeline

Workflow



Example

```
name: Deploy Python project to Azure Function App

on:
  [push]

env:
  AZURE_FUNCTIONAPP_NAME: 'your-app-name' # set this to your function app name on Azure
  AZURE_FUNCTIONAPP_PACKAGE_PATH: '.' # set this to the path to your function app project, default
  PYTHON_VERSION: '3.9' # set this to the python version to use (e.g. '3.6', '3.7',

jobs:
  build-and-deploy:
    runs-on: ubuntu-latest
    environment: dev
    steps:
      - name: 'Checkout GitHub Action'
        uses: actions/checkout@v3

      - name: Setup Python ${ env.PYTHON_VERSION } Environment
        uses: actions/setup-python@v4
        with:
          python-version: ${ env.PYTHON_VERSION }

      - name: 'Resolve Project Dependencies Using Pip'
        shell: bash
        run: |
          pushd './${ env.AZURE_FUNCTIONAPP_PACKAGE_PATH }'
          python -m pip install --upgrade pip
          pip install -r requirements.txt --target=".python_packages/lib/site-packages"
          popd

      - name: 'Run Azure Functions Action'
        uses: Azure/functions-action@v1
        id: fa
        with:
          app-name: ${ env.AZURE_FUNCTIONAPP_NAME }
          package: ${ env.AZURE_FUNCTIONAPP_PACKAGE_PATH }
          publish-profile: ${ secrets.AZURE_FUNCTIONAPP_PUBLISH_PROFILE }
          scm-do-build-during-deployment: true
          enable-oryx-build: true
```

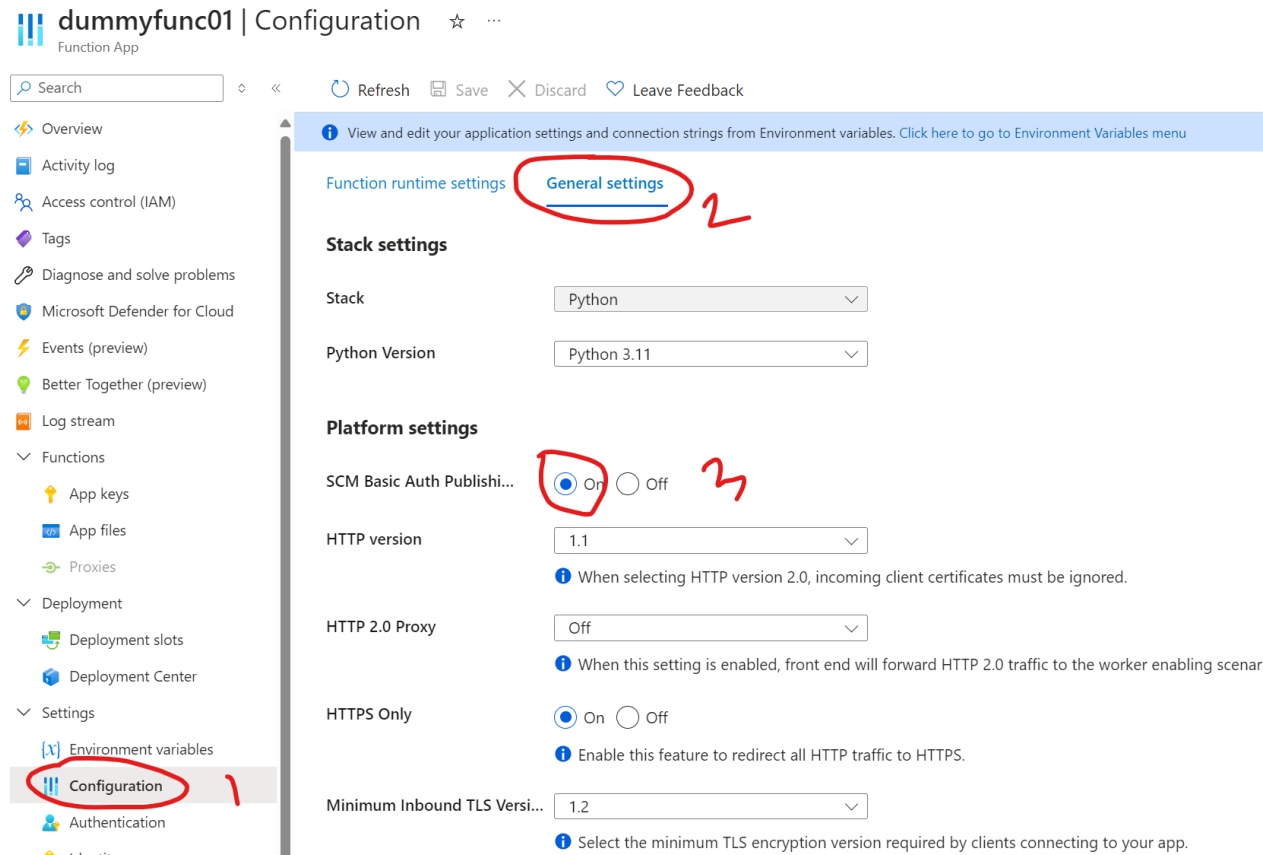
CI/CD Pipeline using GitHub Actions

Demo

Task Three: CI/CD Pipeline

Setup Your Own Pipeline using GitHub Actions

- Fork Python Sample Code from <https://github.com/ruddra/AzureFunction>
- Fix Configuration in Azure Function



Task Three: CI/CD Pipeline

- Download Publish Profile as mentioned here: [Use GitHub Actions to make code updates in Azure Functions | Microsoft Learn](#)
- Set Secret in GitHub Repository [Use GitHub Actions to make code updates in Azure Functions | Microsoft Learn](#)
- Change the Pipeline of the source code to match the function name. (Inside .github/workflows/deploy_func.yml)
- Push the code to see if it runs the action

FinOps

What is FinOps?

FinOps is an operational framework and cultural practice which maximizes the business value of cloud and technology, enables timely data-driven decision making, and creates financial accountability through collaboration between engineering, finance, and business teams.

FinOps Foundation [Technical Advisory Council](#)
Updated: January 2025

FinOps Framework

Principles



Teams need to collaborate



Decisions are driven by business value of cloud



Everyone takes ownership for their cloud usage



FinOps data should be accessible and timely



A centralized team drives FinOps



Take advantage of the variable cost model of the cloud

Domains & Capabilities



Understand Cloud Usage & Cost



Data Ingestion



Allocation



Reporting & Analytics



Anomaly Management

Quantify Business Value



Planning & Estimating



Forecasting



Budgeting



Benchmarking



Unit Economics

Optimize Cloud Usage & Cost



Architecting for Cloud



Workload Optimization



Licensing & SaaS



Rate Optimization



Cloud Sustainability

Manage the FinOps Practice



FinOps Practice Operations



Cloud Policy & Governance



FinOps Assessment



FinOps Tools & Services



FinOps Education & Enablement



Invoicing & Chargeback



Onboarding Workloads



Intersecting Disciplines

Core Personas



Engineering



FinOps Practitioner



Finance



Leadership



Procurement



Product

Allied Personas



ITAM



ITFM



ITSM



Security



Sustainability

Phases



Maturity



Crawl



Walk



Run

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Bonus Materials

House Keeping

- Cloud Shell
- Create Resources using Azure Shell
- Infra Pipeline
- Azure Policy
- Azure Certification Path

Bonus Challenges

- Create an Azure Function locally: <https://learn.microsoft.com/en-us/azure/azure-functions/functions-develop-local>
- Deploy Azure function from local machine: <https://learn.microsoft.com/en-us/azure/azure-functions/deployment-zip-push>
- Deploy using GitHub Actions: <https://learn.microsoft.com/en-us/azure/azure-functions/functions-how-to-github-actions?tabs=linux%2Cdotnet&pivots=method-template>

*Thank you for
your attention*